

Part of the Monistic Continuum Model (MCM)

Physics–MCM Correspondence: Emergence of Physical Structures from Monistic Geometry

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Abstract

This document is the first publication of the four-part series MCM Continuum Physics Series. It introduces the formal correspondence between established physics and the Monistic Continuum Model (MCM). Physics and the MCM do not contradict each other; they operate on different descriptive levels. Physics describes the world after the emergence of space, while the MCM describes the underlying geometric processes before the emergence of space. Physical quantities, fields, particles, and forces correspond to emergent structures of the medium in the MCM. This first document establishes the conceptual and mathematical bridge between both perspectives and forms the foundation for the following publications:

MCM Continuum Physics Series

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+++ 01_Physics--MCM Correspondence

+++ 02_Physics Derived from the MCM (forthcoming)

+++ 03_Observable Phenomena of the MCM (forthcoming)

\-- 04_Foundation of Future MCM Publications (forthcoming)

The series aims not only to clarify the role of the MCM within physics, but also to strengthen the significance of the physical world compared to the relativistic world of physics. Understanding the cosmos requires more than thinking about what has already been achieved; it requires rethinking what is possible. Only by recognizing the limits of our models can we truly comprehend the structure of reality. This document therefore marks the beginning of a broader effort to expand and rethink physics through the lens of monistic geometry.

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1 Introduction

Physics and the MCM are not competing theories. They describe the same reality from different perspectives:

- Physics describes the world as it *appears*.
- The MCM describes the world as it *is*.

Physics works with space, time, fields, and particles. The MCM works with stress, rotation, curvature, and internal updating of the medium.

Both levels are fully compatible.

2 Two Levels of Description

2.1 Physics: Level of Appearance

Physics uses:

- space and time,
- fields (electromagnetic, gravitational),
- particles,
- forces,
- energy and momentum.

These quantities are not fundamental but describe the world *after* the emergence of space.

2.2 MCM: Origin and Structural Level

The MCM uses:

$$C(\alpha, \Omega) = (p, v_A, \sigma_{AB}, \tau),$$

and describes:

- stress as the source of all geometry,
- rotation as an internal geometric quantity,
- internal curvature,
- internal updating,
- vortices as the only stable structure,
- emergence of space from bound geometry.

These quantities are fundamental.

3 The Correspondence

The correspondence between physics and the MCM arises through the emergence mapping:

$$x = X(\alpha, \Omega),$$

which transforms the pre-spatial medium index α into spatial coordinates x . The spatial metric is a projection of the stress-induced medium metric:

$$g_{ij}(x, \Omega) = \gamma_{AB}(\alpha, \Omega) \frac{\partial X^A}{\partial x^i} \frac{\partial X^B}{\partial x^j}.$$

Thus:

Physics = MCM seen from the outside.

MCM = Physics seen from the inside.

4 Correspondence of Physical Quantities

4.1 Mass

Physics: property of a particle. MCM: bound geometry of a vortex.

4.2 Energy

Physics: scalar quantity. MCM: organization of stress in the medium.

4.3 Spacetime

Physics: background structure. MCM: emergent metric from bound geometry.

4.4 Fields

Physics: electromagnetic or gravitational fields. MCM: rotational-geometric patterns of the medium.

4.5 Particles

Physics: point-like objects. MCM: stable vortex structures.

4.6 Forces

Physics: interactions between particles. MCM: stress gradients:

$$Dv_A = G(\sigma, \nabla \sigma).$$

5 Correspondence of Physical Equations

5.1 Einstein Equation

$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

MCM projection:

$$\gamma_{AB} \rightarrow g_{ij}, \quad F(Vv, \sigma) \rightarrow T_{\mu\nu}.$$

5.2 Maxwell Equations

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

MCM projection:

$$L_{AB} \rightarrow F_{\mu\nu}.$$

5.3 Quantization

Physics: fundamental property. MCM: stability modes of bound geometry.

6 Unification

The unification of physics and the MCM is:

$$\text{Physics} = \Pi(C(\alpha, \Omega)),$$

where Π is the projection of the medium onto emergent space.

Thus physics is:

- not wrong,
- not incompatible,
- not contradictory,
- but a different level of description.

7 Glossary of Symbols and Concepts

$C(\alpha, \Omega)$ Pre-spatial state mapping of the medium; contains pressure p , velocity v_A , stress σ_{AB} , and reactivity τ .

$x = X(\alpha, \Omega)$ Emergence mapping; transforms the medium index α into spatial coordinates x .

$g_{ij}(x, \Omega)$ Emergent metric; projection of the medium metric γ_{AB} into observable space.

γ_{AB} Medium metric; internal geometric relation of the medium, formed from stress σ_{AB} .

σ_{AB} Stress tensor of the medium; source of all internal geometry and curvature.

L_{AB} Rotation operator; describes internal geometric rotation of the medium.

R_{ABCD} Curvature operator; defines internal medium curvature.

$Dv_A = G(\sigma, \nabla\sigma)$ Expression for stress gradients; describes geometry-induced motion.

$DC(\alpha, \Omega)$ Updating operator; defines internal dynamics and stability of bound geometry.

Medium The only ontological entity of the MCM; origin of all geometry, rotation, and curvature.

Vortex Stable bound process geometry of the medium; basis of all emergent structures.

Emergence Transition from pre-spatial geometry to observable structure; produces space and physical phenomena.

Correspondence Mapping between MCM geometry and physical concepts (space, time, fields, particles, forces).

Physics Outer appearance of the MCM structure after the emergence of space.

MCM Monistic Continuum Model; describes the internal geometric structure of the medium before the emergence of space.

8 Summary

- Physics describes the world after the emergence of space.
- The MCM describes the world before the emergence of space.
- Physical quantities are emergent MCM structures.
- Physical equations are projections of MCM dynamics.
- Both theories are fully compatible.
- Physics is the image; the MCM is the structure.

9 References

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